SOC-Based Electricity Market Design and Competitive-Equilibrium KKT Derivation

Centralized ISO clearing, conventional generators, standalone storage, and storage hybridised with a variable resource

Technical derivation note

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Abstract

We derive in full the Karush–Kuhn–Tucker (KKT) system of the ISO clearing problem for state-of-charge (SOC) based storage participation, with the network, generation and demand constraints written out explicitly rather than collected into the compact operators Bx and $Cx \le d$; write the equilibrium problem of the market participants as a mixed complementarity problem (MCP); and prove that the two systems coincide. The ISO's dual variable on the SOC balance is shown to be an ISO-computed shadow price of stored energy obeying a backward Bellman recursion whose terminal condition is the end-of-horizon bid, and the nodal prices are shown to decompose into a reference price plus a shift-factor weighted sum of congestion prices. Consequently the ISO dispatch, together with the duals of the nodal balance, is a competitive equilibrium: no participant — including the storage owner, which submits no in-horizon energy offer — has a profitable unilateral deviation, and the settlement is revenue adequate with merchandising surplus exactly equal to congestion rent.

We then extend the model to *hybrid* resources: storage co-located with a variable generator behind a shared interconnection. We show that the entire storage block of the KKT system survives hybridisation under a single substitution, the nodal price $\lambda_{n,t}$ being replaced by an *internal price* $\widetilde{\lambda}_{h,t} = \lambda_{n,t} - \chi_{h,t}$ in which $\chi_{h,t}$ is the multiplier of the interconnection limit. The Bellman recursion, the no-simultaneous-charge-discharge result, individual rationality and revenue adequacy all carry over verbatim; the arbitrage condition does not. When the hybrid curtails, $\chi_{h,t}$ absorbs $\lambda_{n,t}$ exactly and the nodal price disappears from the storage's first-order condition, which then reduces to a pure degradation threshold. Four conditions are shown to be necessary for the equilibrium result — concavity of the end-of-horizon bid, convexity of the feasible set, separability of every non-clearing constraint by ownership, and congruence between the constraint set the ISO imposes and the one the owner faces — and hybridisation is the first construct in the paper that violates the third and fourth while satisfying the first two. We verify the whole system numerically on four cases: the single-node six-interval base case of the source paper, where the published price vector turns out not to be dual feasible; a three-node, four-interval networked case with congestion, generator ramping and elastic demand; a hybrid case in which the internal price is zero in two intervals while the nodal price is \$40/MWh; and a case quantifying the welfare cost of a no-grid-charging rule. A self-scheduled hybrid with a mis-forecast peak loses 27.0% of welfare in the same instance.

1 Scope

References to the source paper [1] are written as "Eq. (4) of [1]"; all displayed equations of this note are numbered independently in the right margin. Problem (9) of [1] is stated compactly as

$$\min_{x,p,g,s} \Big\{ \Delta T c^\top x + \Delta T(\epsilon 1^\top p + \zeta 1^\top g) - b(s_T - \bar{s}_T) \;\Big|\; (1)\text{–}(7),\ p - g = Bx,\ Cx \le d \Big\}, \qquad (1)$$

in which the entire non-storage market — network, generators, loads — is compressed into the injection map B and the polyhedron $Cx \le d$. That compression is harmless for the existence

argument but it hides exactly the objects a market model has to price: nodal balance, Kirchhoff's laws, thermal limits, offer blocks, ramp rates, and demand bids. Section 3 therefore expands (1) into its full disaggregated form, and everything that follows is derived from the expanded model. Section 3.1 records the mapping back to (1).

The compression hides one further object, and it is the one that has grown fastest in interconnection queues: the shared point of interconnection of a *hybrid* resource. Writing $p - g = Bx$ presumes that the storage's injection is limited only by its own power rating and by the network. A hybrid's injection is limited by a third thing — the capacity it shares with the generator it is co-located with — and that third thing is neither a network constraint (it is not in $Cx \le d$, and no ISO computes a shift factor for it) nor a private storage constraint (it couples the storage to another machine). Section 8 adds it, and shows that this single affine constraint is enough to change the storage's marginal valuation of energy from a market quantity into a private one.

2 Notation

Sets and indices			
$t \in \mathcal{T} = \{1, \dots, T\}$	dispatch intervals, length ΔT (h)	$n \in \mathcal{N}$	buses (nodes)
$\ell \in \mathcal{L}$	transmission lines	$i \in \mathcal{G}$, $k \in \mathcal{K}_i$	generating units, offer blocks
$m \in \mathcal{D}$, $j \in \mathcal{J}_m$	loads, demand-bid blocks	$e \in \mathcal{E}$	standalone storage units
$h \in \mathcal{H}$	hybrid resources	$\mathcal{S} \subseteq \mathcal{H}$	set sharing one interconnection
$\mathcal{G}(n), \mathcal{D}(n), \mathcal{E}(n), \mathcal{H}(n)$	units at bus n	$n(i), n(m), n(e), n(h)$	bus of a unit
Primal variables			
$q_{i,k,t} \ge 0$	generation from block k (MW)	$d_{m,j,t} \ge 0$	served demand block (MW)
$f_{\ell,t}$	active power flow on line ℓ (MW)	$\theta_{n,t}$	voltage angle at bus n (rad)
$p_{e,t} \ge 0$	storage charge power (MW)	$g_{e,t} \ge 0$	storage discharge power (MW)
$s_{e,t}$	SOC at end of interval t (MWh)	$r_{h,t} \ge 0$	dispatched output of the hybrid plant (MW)
$z_{h,t}$	hybrid net injection at the POI (MW)		
Data			
$c_{i,k}$	offer price of block k (\$/MWh)	$\bar{q}_{i,k}$	block size (MW)
$u_{m,j}$	demand bid price (\$/MWh)	$\bar{d}_{m,j}$	bid block size (MW)
$\bar{R}_i, \underline{R}_i$	up / down ramp limits (MW/interval)	y_ℓ	series susceptance (MW/rad)
$\bar{f}_\ell$	thermal limit (MW)	$A \in \mathbb{R}^{\lvert\mathcal{N}\rvert \times \lvert\mathcal{L}\rvert}$	bus–line incidence matrix
α_e, β_e	charge / inverse-discharge efficiency	γ_e	self-discharge retention factor
ϵ_e, ζ_e	charge / discharge degradation (\$/MWh)	$\bar{s}_e, \underline{s}_e$	SOC limits (MWh)
$\bar{p}_e, \bar{g}_e$	power limits (MW)	$b_e(\cdot), w_e(\cdot)$	end-of-horizon bid and its slope
$D_{n,t}$	fixed (price-inelastic) demand (MW)	$r \in \mathcal{N}$	reference bus
$R_{h,t}$	availability of the hybrid plant (MW)	c_h	offer price of that plant (\$/MWh)
$\bar{P}_h$, $\underline{P}_h$	POI injection / withdrawal limit (MW)	Θ_h	allowance for grid charging (MW)
Multipliers introduced by hybridisation (all in \$/MWh)			
$\varphi_{h,t}$, $\underline{\varphi}_{h,t} \ge 0$	bounds on $r_{h,t}$	$\underline{\chi}_{h,t}$, $\underline{\chi}_{h,t} \ge 0$	POI injection / withdrawal limit
$\omega_{h,t} \ge 0$	no-grid-charging rule	$\widetilde{\lambda}_{h,t}$	internal price, Eq. (31)

$A_{n\ell} = +1$ if n is the from-bus of ℓ, -1 if the to-bus, 0 otherwise; write $\mathrm{fr}(\ell)$ and $\mathrm{to}(\ell)$ for those buses. The storage recursion, Eq. (4) of [1], is

$$s_{e,t} = \gamma_e s_{e,t-1} + \Delta T\, \alpha_e p_{e,t} - \Delta T\, \beta_e g_{e,t}, \qquad e \in \mathcal{E},\ t \in \mathcal{T}, \tag{2}$$

with $s_{e,0}$ given, and the tightened SOC limits, Eqs. (6)–(7) of [1], are

$$\gamma_e s_{e,t-1} + \Delta T\, \alpha_e p_{e,t} \le \bar{s}_e, \qquad \underline{s}_e - \gamma_e s_{e,t-1} + \Delta T\, \beta_e g_{e,t} \le 0. \tag{3}$$

Substituting (2) into (3) gives the equivalent readable forms $s_{e,t} + \Delta T \beta_e g_{e,t} \le \bar{s}_e$ and $s_{e,t} - \Delta T \alpha_e p_{e,t} \ge \underline{s}_e$, which is why (3) is tighter than bounding $s_{e,t}$ alone: it forbids trajectories that respect the limits at interval endpoints but violate them within the interval. The end-of-horizon bid, Eq. (8) of [1], is

$$b_e(\Delta) = \int_0^{\Delta} w_e(\sigma)\, \mathrm{d}\sigma, \qquad \Delta := s_{e,T} - \bar{s}_{e,T}, \tag{4}$$

with $\bar{s}_{e,T}$ the reference terminal SOC.

Assumption 1 (Bid validity)**.** w_e is non-increasing for every $e \in \mathcal{E}$. Hence each b_e is concave and $-b_e$ convex.

Assumption 2 (Convex market)**.** The mutual-exclusivity condition, Eq. (3) of [1], is relaxed; no binary variables are present (Appendices A.2, A.3 of [1] excluded); losses are neglected or piecewise-linearised convexly; offer stacks are monotone, $c_{i,1} \le c_{i,2} \le \cdots$, and demand stacks are monotone non-increasing, $u_{m,1} \ge u_{m,2} \ge \cdots$.

Assumption 3 (Ownership separability)**.** Every constraint of the ISO problem other than the nodal balance involves only variables belonging to a single commercial agent (or only to the network).

Assumption 4 (Model congruence)**.** For every agent, the feasible set the ISO imposes in its clearing problem coincides with the feasible set the agent actually faces, and the prices used in settlement are duals of that same clearing problem.

Assumptions 1 and 2 are logically independent: the first concerns the *curvature of a function*, the second the *shape of a set*. Assumptions 3 and 4 are implicit throughout the source paper and are usually left unstated because for thermal units, loads and standalone storage they are automatic: a ramp rate belongs to one machine and an ISO that models a ramp rate models the one the unit has. They are stated here because they are exactly what hybridisation puts at risk, and because they fail in ways that convexity arguments cannot detect (Sections 8.7–8.8 and Section 10).

3 The ISO problem in disaggregated form

$$\min_{q,d,f,\theta,p,g,s} \quad \Delta T \sum_{t\in\mathcal{T}} \Bigg[\sum_{i\in\mathcal{G}} \sum_{k\in\mathcal{K}_i} c_{i,k}\, q_{i,k,t} - \sum_{m\in\mathcal{D}} \sum_{j\in\mathcal{J}_m} u_{m,j}\, d_{m,j,t} + \sum_{e\in\mathcal{E}} \big(\epsilon_e p_{e,t} + \zeta_e g_{e,t}\big) \Bigg] - \sum_{e\in\mathcal{E}} b_e\big(s_{e,T} - \bar{s}_{e,T}\big) \tag{5}$$

subject to, for all $t \in \mathcal{T}$, with multipliers on the right:

$$\sum_{m\in\mathcal{D}(n)} \sum_{j} d_{m,j,t} + D_{n,t} + \sum_{\ell\in\mathcal{L}} A_{n\ell} f_{\ell,t} - \sum_{i\in\mathcal{G}(n)} \sum_{k} q_{i,k,t} - \sum_{e\in\mathcal{E}(n)} \big(g_{e,t} - p_{e,t}\big) = 0, \qquad \Delta T\lambda_{n,t} \text{ free} \tag{D1}$$

$$y_\ell\big(\theta_{\mathrm{fr}(\ell),t} - \theta_{\mathrm{to}(\ell),t}\big) - f_{\ell,t} = 0, \qquad \Delta T\kappa_{\ell,t} \text{ free} \tag{D2}$$

$$f_{\ell,t} - \bar{f}_\ell \le 0, \qquad -\bar{f}_\ell - f_{\ell,t} \le 0, \qquad \Delta T\bar{\mu}_{\ell,t},\ \Delta T\underline{\mu}_{\ell,t} \ge 0 \tag{D3}$$

$$\theta_{r,t} = 0, \qquad \Delta T\xi_t \text{ free} \tag{D4}$$

$$-q_{i,k,t} \le 0, \qquad q_{i,k,t} - \bar{q}_{i,k} \le 0, \qquad \Delta T\underline{\delta}_{i,k,t},\ \Delta T\bar{\delta}_{i,k,t} \ge 0 \tag{D5}$$

$$\sum_k q_{i,k,t} - \sum_k q_{i,k,t-1} - \bar{R}_i \le 0, \qquad \Delta T\bar{\tau}_{i,t} \ge 0 \tag{D6}$$

$$\sum_k q_{i,k,t-1} - \sum_k q_{i,k,t} - \underline{R}_i \le 0, \qquad \Delta T\underline{\tau}_{i,t} \ge 0 \tag{D7}$$

$$-d_{m,j,t} \le 0, \qquad d_{m,j,t} - \bar{d}_{m,j} \le 0, \qquad \Delta T\underline{\psi}_{m,j,t},\ \Delta T\bar{\psi}_{m,j,t} \ge 0 \tag{D8}$$

$$s_{e,t} - \gamma_e s_{e,t-1} - \Delta T\alpha_e p_{e,t} + \Delta T\beta_e g_{e,t} = 0, \qquad \pi_{e,t} \text{ free} \tag{D9}$$

$$\gamma_e s_{e,t-1} + \Delta T\alpha_e p_{e,t} - \bar{s}_e \le 0, \qquad \Delta T\bar{\nu}_{e,t} \ge 0 \tag{D10}$$

$$\underline{s}_e - \gamma_e s_{e,t-1} + \Delta T \beta_e g_{e,t} \le 0, \qquad \Delta T \underline{\nu}_{e,t} \ge 0 \tag{D11}$$

$$-p_{e,t} \le 0, \quad p_{e,t} - \bar{p}_e \le 0, \qquad \Delta T \underline{\rho}_{e,t},\ \Delta T \bar{\rho}_{e,t} \ge 0 \tag{D12}$$

$$-g_{e,t} \le 0, \quad g_{e,t} - \bar{g}_e \le 0, \qquad \Delta T \underline{\sigma}_{e,t},\ \Delta T \bar{\sigma}_{e,t} \ge 0 \tag{D13}$$

for all $n \in \mathcal{N}$, $\ell \in \mathcal{L}$, $i \in \mathcal{G}$, $k \in \mathcal{K}_i$, $m \in \mathcal{D}$, $j \in \mathcal{J}_m$, $e \in \mathcal{E}$; (D6)–(D7) for $t \ge 2$.

Three modelling points. First, (D1) is written as *net withdrawal* $= 0$ so that $\lambda_{n,t}$ comes out as a price rather than its negative; the factors ΔT are inserted so that every multiplier carries units of \$/MWh. Second, (D2) is the DC linearisation of Kirchhoff's voltage law, and it is what makes the network more than a transportation problem: without it, flows could be routed freely and the price decomposition of Proposition 1 would collapse. Third, (D6)–(D7) give thermal units an *intertemporal* constraint of their own, so the comparison between the storage costate and the ramp wedge in Section 5 is available.

3.1 Mapping back to the compact form

Setting $x := \big(q, d, f, \theta\big)$, the compact objects of (1) are

$$c^\top x := \sum_{i,k,t} c_{i,k} q_{i,k,t} - \sum_{m,j,t} u_{m,j} d_{m,j,t}, \qquad (Bx)_{n,t} := \sum_{i \in \mathcal{G}(n),k} q_{i,k,t} - \sum_{m \in \mathcal{D}(n),j} d_{m,j,t} - D_{n,t} - \sum_{\ell} A_{n\ell} f_{\ell,t}, \tag{6}$$

so that (D1) is $p_{n,t} - g_{n,t} = (Bx)_{n,t}$ aggregated over $\mathcal{E}(n)$, and $Cx \le d$ collects (D3) and (D5)–(D8), with (D2) and (D4) as the equality rows of the same block. In the notation of [1] the multiplier μ on $Cx \le d$ is $\mu = \big(\bar{\mu}, \underline{\mu}, \bar{\delta}, \underline{\delta}, \bar{\tau}, \underline{\tau}, \bar{\psi}, \underline{\psi}\big)$, of which $\mu_s := (\bar{\mu}, \underline{\mu})$ are the system-wide (transmission) components and the remainder are firm-private. Hybridisation, treated in Section 8, adds a third category that the compact form has no slot for: multipliers that are firm-private but attach to a *shared physical asset*.

4 KKT system of the ISO problem

The Lagrangian is

$$\begin{aligned}
\mathcal{L} = (5) &+ \Delta T \sum_{n,t} \lambda_{n,t} \Big[\textstyle\sum_{m,j} d_{m,j,t} + D_{n,t} + \sum_{\ell} A_{n\ell} f_{\ell,t} - \sum_{i,k} q_{i,k,t} - \sum_{e} (g_{e,t} - p_{e,t}) \Big] \\
&+ \Delta T \sum_{\ell,t} \kappa_{\ell,t} \Big[y_\ell \big(\theta_{\mathrm{fr}(\ell),t} - \theta_{\mathrm{to}(\ell),t} \big) - f_{\ell,t} \Big] + \Delta T \sum_{\ell,t} \Big[\bar{\mu}_{\ell,t} (f_{\ell,t} - \bar{f}_\ell) + \underline{\mu}_{\ell,t} (-\underline{f}_\ell - f_{\ell,t}) \Big] \\
&+ \Delta T \sum_{t} \xi_t \theta_{r,t} + \Delta T \sum_{i,k,t} \big[\bar{\delta}_{i,k,t} (q_{i,k,t} - \bar{q}_{i,k}) - \underline{\delta}_{i,k,t} q_{i,k,t} \big] \\
&+ \Delta T \sum_{i,t \ge 2} \Big[\bar{\tau}_{i,t} \big(Q_{i,t} - Q_{i,t-1} - \bar{R}_i \big) + \underline{\tau}_{i,t} \big(Q_{i,t-1} - Q_{i,t} - \underline{R}_i \big) \Big] \\
&+ \Delta T \sum_{m,j,t} \big[\bar{\psi}_{m,j,t} (d_{m,j,t} - \bar{d}_{m,j}) - \underline{\psi}_{m,j,t} d_{m,j,t} \big] \ + \ \mathcal{L}^{\mathrm{stor}},
\end{aligned} \tag{7}$$

where $Q_{i,t} := \sum_k q_{i,k,t}$ and

$$\begin{aligned}
\mathcal{L}^{\mathrm{stor}} = \sum_{e,t} \Big\{ &\pi_{e,t} \big[s_{e,t} - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} + \Delta T \beta_e g_{e,t} \big] + \Delta T \bar{\nu}_{e,t} \big[\gamma_e s_{e,t-1} + \Delta T \alpha_e p_{e,t} - \bar{s}_e \big] \\
&+ \Delta T \underline{\nu}_{e,t} \big[\underline{s}_e - \gamma_e s_{e,t-1} + \Delta T \beta_e g_{e,t} \big] + \Delta T \big[\bar{\rho}_{e,t} (p_{e,t} - \bar{p}_e) - \underline{\rho}_{e,t} p_{e,t} \\
&+ \bar{\sigma}_{e,t} (g_{e,t} - \bar{g}_e) - \underline{\sigma}_{e,t} g_{e,t} \big] \Big\}.
\end{aligned} \tag{8}$$

The multiplier $\pi_{e,t}$ on (D9) is the *marginal value of stored energy* at the end of interval t (\$/MWh of SOC); it does all the economic work below.

4.1 Stationarity

Generation $q_{i,k,t}$. $\partial\mathcal{L}/\partial q_{i,k,t} = \Delta T\big[c_{i,k} - \lambda_{n(i),t} + \bar{\delta}_{i,k,t} - \underline{\delta}_{i,k,t} + (\bar{\tau}_{i,t} - \underline{\tau}_{i,t}) - (\bar{\tau}_{i,t+1} - \underline{\tau}_{i,t+1})\big] = 0$, i.e.

$$\lambda_{n(i),t} = c_{i,k} + \bar{\delta}_{i,k,t} - \underline{\delta}_{i,k,t} + \underbrace{(\bar{\tau}_{i,t} - \underline{\tau}_{i,t}) - (\bar{\tau}_{i,t+1} - \underline{\tau}_{i,t+1})}_{=:\,\varrho_{i,t}\ \text{(ramp wedge)}}, \tag{K1a}$$

with the convention $\bar{\tau}_{i,1} = \underline{\tau}_{i,1} = \bar{\tau}_{i,T+1} = \underline{\tau}_{i,T+1} = 0$.

Demand $d_{m,j,t}$. $\partial\mathcal{L}/\partial d_{m,j,t} = \Delta T\big[-u_{m,j} + \lambda_{n(m),t} + \bar{\psi}_{m,j,t} - \underline{\psi}_{m,j,t}\big] = 0$, i.e.

$$\lambda_{n(m),t} = u_{m,j} - \bar{\psi}_{m,j,t} + \underline{\psi}_{m,j,t}. \tag{K1b}$$

Flows $f_{\ell,t}$. The term $A^\top\lambda$ enters from (D1):

$$\kappa_{\ell,t} = \big(\lambda_{\mathrm{fr}(\ell),t} - \lambda_{\mathrm{to}(\ell),t}\big) + \big(\bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t}\big) = \big(A^\top\lambda_t\big)_\ell + \nu_{\ell,t}, \qquad \nu_{\ell,t} := \bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t}. \tag{K1c}$$

Angles $\theta_{n,t}$. Only (D2) and (D4) contain θ:

$$\big(A\,\mathrm{diag}(y)\,\kappa_t\big)_n + \xi_t\,\mathbf{1}\{n = r\} = 0 \qquad \Longleftrightarrow \qquad A\,\mathrm{diag}(y)\,\kappa_t = -\xi_t e_r. \tag{K1d}$$

Summing (K1d) over n and using $\mathbf{1}^\top A = 0$ gives $\xi_t = 0$: the reference angle is a normalisation and carries no economic content. Hence $A\,\mathrm{diag}(y)\,\kappa_t = 0$, which is the dual form of Kirchhoff's voltage law.

Charging $p_{e,t}$.

$$\lambda_{n(e),t} + \epsilon_e = \alpha_e\big(\pi_{e,t} - \bar{\nu}_{e,t}\big) + \underline{\rho}_{e,t} - \bar{\rho}_{e,t}. \tag{K2}$$

Discharging $g_{e,t}$.

$$\lambda_{n(e),t} = \zeta_e + \beta_e\big(\pi_{e,t} + \underline{\nu}_{e,t}\big) - \underline{\sigma}_{e,t} + \bar{\sigma}_{e,t}. \tag{K3}$$

SOC $s_{e,t}$, $t = 1, \dots, T-1$.

$$\pi_{e,t} = \gamma_e\big(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\big). \tag{K4}$$

Terminal SOC $s_{e,T}$.

$$\pi_{e,T} = w_e\big(s_{e,T} - \bar{s}_{e,T}\big), \qquad \text{or} \quad \pi_{e,T} \in \partial b_e\big(s_{e,T} - \bar{s}_{e,T}\big) \text{ for piecewise-linear } b_e. \tag{K5}$$

4.2 Dual feasibility and complementary slackness

All of $\bar{\mu}, \underline{\mu}, \bar{\delta}, \underline{\delta}, \bar{\tau}, \underline{\tau}, \bar{\psi}, \underline{\psi}, \bar{\nu}, \underline{\nu}, \bar{\rho}, \underline{\rho}, \bar{\sigma}, \underline{\sigma} \geq 0$ (K6), and

$$0 \leq \bar{\mu}_{\ell,t} \ \perp\ \bar{f}_\ell - f_{\ell,t} \geq 0, \qquad 0 \leq \underline{\mu}_{\ell,t} \ \perp\ f_{\ell,t} + \bar{f}_\ell \geq 0, \tag{K7}$$

$$0 \leq \underline{\delta}_{i,k,t} \ \perp\ q_{i,k,t} \geq 0, \qquad 0 \leq \bar{\delta}_{i,k,t} \ \perp\ \bar{q}_{i,k} - q_{i,k,t} \geq 0, \tag{K8}$$

$$0 \leq \bar{\tau}_{i,t} \ \perp\ \bar{R}_i - Q_{i,t} + Q_{i,t-1} \geq 0, \qquad 0 \leq \underline{\tau}_{i,t} \ \perp\ \underline{R}_i - Q_{i,t-1} + Q_{i,t} \geq 0, \tag{K9}$$

$$0 \leq \underline{\psi}_{m,j,t} \ \perp\ d_{m,j,t} \geq 0, \qquad 0 \leq \bar{\psi}_{m,j,t} \ \perp\ \bar{d}_{m,j} - d_{m,j,t} \geq 0, \tag{K10}$$

$$0 \leq \bar{\nu}_{e,t} \ \perp\ \bar{s}_e - \gamma_e s_{e,t-1} - \Delta T\alpha_e p_{e,t} \geq 0, \qquad 0 \leq \underline{\nu}_{e,t} \ \perp\ \gamma_e s_{e,t-1} - \Delta T\beta_e g_{e,t} - \underline{s}_e \geq 0, \tag{K11}$$

$$0 \leq \underline{\rho}_{e,t} \ \perp\ p_{e,t} \geq 0, \qquad 0 \leq \bar{\rho}_{e,t} \ \perp\ \bar{p}_e - p_{e,t} \geq 0,$$

$$0 \leq \underline{\sigma}_{e,t} \ \perp\ g_{e,t} \geq 0, \qquad 0 \leq \bar{\sigma}_{e,t} \ \perp\ \bar{g}_e - g_{e,t} \geq 0, \tag{K12}$$

together with primal feasibility (D1)–(D13). Here $a \perp b$ means $a^\top b = 0$.

4.3 Reduced form: eliminating the bound multipliers

Since $\underline{\rho}_{e,t}$ and $\bar{\rho}_{e,t}$ cannot both be positive, (K2) with (K12) is equivalent to the *box complementarity* statement

$$0 \le p_{e,t} \le \bar{p}_e \quad \perp \quad F^p_{e,t} := \lambda_{n(e),t} + \epsilon_e - \alpha_e\big(\pi_{e,t} - \bar{\nu}_{e,t}\big), \tag{K2$'$}$$

read as

$$p_{e,t} = 0 \Rightarrow F^p_{e,t} \ge 0, \qquad 0 < p_{e,t} < \bar{p}_e \Rightarrow F^p_{e,t} = 0, \qquad p_{e,t} = \bar{p}_e \Rightarrow F^p_{e,t} \le 0, \tag{9}$$

and likewise

$$0 \le g_{e,t} \le \bar{g}_e \;\perp\; F^g_{e,t} := \zeta_e + \beta_e\big(\pi_{e,t} + \underline{\nu}_{e,t}\big) - \lambda_{n(e),t}, \qquad 0 \le q_{i,k,t} \le \bar{q}_{i,k} \;\perp\; c_{i,k} + \varrho_{i,t} - \lambda_{n(i),t}, \tag{K3$'$}$$

$$0 \le d_{m,j,t} \le \bar{d}_{m,j} \;\perp\; \lambda_{n(m),t} - u_{m,j}, \qquad -\bar{f}_\ell \le f_{\ell,t} \le \bar{f}_\ell \;\perp\; \kappa_{\ell,t} - \big(A^\top \lambda_t\big)_\ell. \tag{K1$'$}$$

The middle case of (9) is the operative one for pricing: *whenever a resource operates strictly interior, its own stationarity condition pins λ at its bus exactly*, leaving no clearing range. Section 9 exploits this, and Section 8.7 shows that a self-scheduled hybrid destroys it.

4.4 Nodal price decomposition

Proposition 1 (LMP decomposition). *Let $B_{\text{bus}} := A\,\mathrm{diag}(y)A^\top$, let $\mathcal{N}_{-r} := \mathcal{N} \setminus \{r\}$, and let the shift-factor (PTDF) matrix relative to reference bus r be $\Psi := \mathrm{diag}(y)\, A_{-r}^\top B_{-r}^{-1}$, where subscript $-r$ deletes the row and column of r. Then* (K1c)–(K1d) *imply*

$$\lambda_{n,t} = \lambda_{r,t} - \sum_{\ell \in \mathcal{L}} \Psi_{\ell n}\big(\bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t}\big), \qquad n \in \mathcal{N}. \tag{10}$$

Proof. Substituting (K1c) into (K1d) with $\xi_t = 0$ gives $A\,\mathrm{diag}(y)\big(A^\top\lambda_t + \nu_t\big) = 0$, i.e. $B_{\text{bus}}\lambda_t = -A\,\mathrm{diag}(y)\,\nu_t$. Since $B_{\text{bus}}\mathbf{1} = 0$, write $\lambda_t = \lambda_{r,t}\mathbf{1} + \tilde{\lambda}_t$ with $\tilde{\lambda}_{r,t} = 0$; deleting row r leaves $B_{-r}\tilde{\lambda}_{-r,t} = -A_{-r}\mathrm{diag}(y)\nu_t$, hence $\tilde{\lambda}_{-r,t} = -\big(\mathrm{diag}(y)A_{-r}^\top B_{-r}^{-1}\big)^\top \nu_t = -\Psi^\top \nu_t$, which is (10). □

So the energy component of every LMP is the reference-bus price, common to all buses and to all storage units, while the congestion component is the only source of locational difference. Storage at a congested bus therefore faces a price that already contains the network's scarcity, and (K2)–(K5) are stated at that price; no separate congestion term appears in the storage conditions. Section 8.4 shows that this is precisely what fails for a hybrid: the interconnection scarcity is *not* in λ, so a second term does appear.

5 Economic content

5.1 The SOC price is a Bellman costate

Conditions (K4)–(K5) form a backward recursion whose terminal condition is supplied by the participant's bid and which the ISO propagates backwards:

$$\pi_{e,T} = w_e(s_{e,T} - \bar{s}_{e,T}), \qquad \pi_{e,t} = \gamma_e\big(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\big), \quad t = T-1, \dots, 1. \tag{11}$$

This is the discrete costate (adjoint) equation of the storage dynamics, the exact analogue of a Bellman recursion for a value-to-go. Each term reads: γ_e discounts stored value for self-discharge; $\bar{\nu}_{e,t+1} > 0$ (tank full) *reduces* $\pi_{e,t}$, since energy carried forward cannot be stored; $\underline{\nu}_{e,t+1} > 0$ (tank empty) *raises* it, since energy carried forward relieves a binding floor. Solving forward from T,

$$\pi_{e,t} = \gamma_e^{T-t}\, w_e(s_{e,T} - \bar{s}_{e,T}) + \sum_{k=t+1}^{T} \gamma_e^{k-t}\big(\underline{\nu}_{e,k} - \bar{\nu}_{e,k}\big). \tag{12}$$

No forecast of λ by the storage owner appears anywhere: the entire intertemporal opportunity cost is reconstructed by the ISO from the single scalar function w_e.

5.2 The ramp wedge is the thermal analogue — and it is weaker

Condition (K1a) contains the corresponding object for a thermal unit, $\varrho_{i,t} = (\bar{\tau}_{i,t} - \underline{\tau}_{i,t}) - (\bar{\tau}_{i,t+1} - \underline{\tau}_{i,t+1})$. It is a *first difference* of ramp multipliers: it couples t only to $t \pm 1$, and it decays to zero the moment no ramp constraint binds. The storage costate (12) is a γ_e-discounted *sum* over the whole remaining horizon and is generically non-zero even when no SOC bound binds, because it inherits w_e at T. That is the formal sense in which energy-limited resources are intertemporal in a way ramp-limited ones are not, and it is why storage needs a terminal bid whereas a thermal unit does not.

5.3 Arbitrage conditions

If storage e charges strictly interior in t and discharges strictly interior in $t' > t$, then (K2′)–(K3′) give $\lambda_{n(e),t} + \epsilon_e = \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t})$ and $\lambda_{n(e),t'} = \zeta_e + \beta_e(\pi_{e,t'} + \underline{\nu}_{e,t'})$. Eliminating π with (11),

$$\boxed{\frac{\lambda_{n(e),t} + \epsilon_e}{\alpha_e} + \bar{\nu}_{e,t} = \gamma_e^{\,t'-t}\,\frac{\lambda_{n(e),t'} - \zeta_e}{\beta_e} + \sum_{k=t+1}^{t'-1} \gamma_e^{\,k-t}\big(\underline{\nu}_{e,k} - \bar{\nu}_{e,k}\big) - \gamma_e^{\,t'-t}\,\bar{\nu}_{e,t'}} \tag{13}$$

If no SOC bound binds on $(t, t']$ and $\gamma_e = 1$, (13) collapses to

$$\lambda_{n(e),t'} = \frac{\beta_e}{\alpha_e}\big(\lambda_{n(e),t} + \epsilon_e\big) + \zeta_e = \frac{1}{\eta^e_{\mathrm{RT}}}\big(\lambda_{n(e),t} + \epsilon_e\big) + \zeta_e : \tag{14}$$

the discharge price must exceed the charge price by the round-trip efficiency factor plus both degradation terms. Since (14) with $t = t'$ is infeasible for $\epsilon_e, \zeta_e \geq 0$ and $\beta_e \geq \alpha_e$, simultaneous charge and discharge never occurs at equilibrium prices, which is why Eq. (3) of [1] can be relaxed.

6 The participants' problems as a complementarity problem

6.1 Lagrangian decomposition

Dualise only the nodal balance (D1) in (7). The remainder separates additively across agents:

$$\mathcal{L} = \sum_{i \in \mathcal{G}} \mathcal{L}_i + \sum_{m \in \mathcal{D}} \mathcal{L}_m + \sum_{e \in \mathcal{E}} \mathcal{L}_e + \mathcal{L}_{\mathrm{net}} + \Delta T \sum_{n,t} \lambda_{n,t} D_{n,t}, \tag{15}$$

because — and this is Assumption 3 — no constraint other than (D1) couples variables belonging to different agents: (D5)–(D7) involve only unit i, (D8) only load m, (D9)–(D13) only storage e, and (D2)–(D4) only the network variables (f, θ). Each block is, up to sign, the objective of one agent's own profit maximisation at price λ. This is the whole of the equivalence result; the rest is bookkeeping.

6.2 Generator i

$$\max_{q_{i,\cdot,\cdot} \geq 0} \; \Delta T \sum_{t \in \mathcal{T}} \Big[\lambda_{n(i),t} Q_{i,t} - \sum_k c_{i,k} q_{i,k,t}\Big] \quad \text{s.t. } q_{i,k,t} \leq \bar{q}_{i,k}, \;\; \big|Q_{i,t} - Q_{i,t-1}\big| \leq \{\bar{R}_i, \underline{R}_i\}, \tag{16}$$

whose KKT system is (K1a) with (K8)–(K9). Note that ramping makes this a genuinely multi-interval problem, so the generator is a price taker *over the whole horizon*, exactly as the storage is.

6.3 Load m

$$\max_{0 \leq d_{m,j,t} \leq \bar{d}_{m,j}} \; \Delta T \sum_{t \in \mathcal{T}} \sum_j \big(u_{m,j} - \lambda_{n(m),t}\big) d_{m,j,t}, \tag{17}$$

with KKT system (K1b) and (K10).

6.4 Storage owner e (price taker)

$$\max_{p_e, g_e, s_e} \ \Delta T \sum_{t\in\mathcal{T}} \Big[\lambda_{n(e),t}\big(g_{e,t} - p_{e,t}\big) - \epsilon_e p_{e,t} - \zeta_e g_{e,t}\Big] + b_e\big(s_{e,T} - \bar{s}_{e,T}\big) \tag{18}$$

subject to the SOC dynamics (2), the SOC limits (3), and $0 \le p_{e,t} \le \bar{p}_e$, $0 \le g_{e,t} \le \bar{g}_e$. The objective is discharge revenue, less charging payment, less degradation, plus the declared value of the terminal SOC position. Attaching the same multiplier names to the same constraints, its KKT system is (K2), (K3), (K4), (K5) with (K11)–(K12), *verbatim*.

6.5 The network

The variables (f, θ) belong to no commercial participant; in an ISO market the system operator holds them. They can nevertheless be assigned to a fictitious price-taking *merchant transmission operator* that buys at each from-bus and sells at each to-bus,

$$\max_{f,\theta} \ -\Delta T \sum_{t\in\mathcal{T}} f_t^\top A^\top \lambda_t \quad \text{s.t.} \quad f_{\ell,t} = y_\ell\big(\theta_{\mathrm{fr}(\ell),t} - \theta_{\mathrm{to}(\ell),t}\big), \quad |f_{\ell,t}| \le \bar{f}_\ell, \quad \theta_{r,t} = 0, \tag{19}$$

whose stationarity conditions in f and θ are exactly (K1c) and (K1d). This is a modelling device, not an institutional claim: the point is that the network block *does* decentralise, but only for an agent constrained to physically consistent flows. An agent that could route power freely subject to thermal limits alone would satisfy (K1c) but not (K1d), and the equivalence would fail — which is the formal statement of why loop flow cannot be traded bilaterally. Section 8.8 constructs the same device for a shared interconnection and shows where it stops working.

6.6 The multi-agent complementarity problem

Definition 2 (MCP)**.** Find $z = (q, d, f, \theta, p, g, s; \lambda, \kappa, \pi, \bar{\mu}, \underline{\mu}, \bar{\delta}, \underline{\delta}, \bar{\tau}, \underline{\tau}, \bar{\psi}, \underline{\psi}, \bar{\nu}, \underline{\nu})$ such that, for all $t \in \mathcal{T}, n \in \mathcal{N}, \ell \in \mathcal{L}, i \in \mathcal{G}, k, m \in \mathcal{D}, j, e \in \mathcal{E}$:

(a) Generators:	$0 \le q_{i,k,t} \le \bar{q}_{i,k} \ \perp \ c_{i,k} + \varrho_{i,t} - \lambda_{n(i),t}$, with (K9)	(20)
(b) Loads:	$0 \le d_{m,j,t} \le \bar{d}_{m,j} \ \perp \ \lambda_{n(m),t} - u_{m,j}$	(21)
(c) Storage, charge:	$0 \le p_{e,t} \le \bar{p}_e \ \perp \ \lambda_{n(e),t} + \epsilon_e - \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t})$	(22)
(d) Storage, discharge:	$0 \le g_{e,t} \le \bar{g}_e \ \perp \ \zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}) - \lambda_{n(e),t}$	(23)
(e) Storage, SOC:	$\pi_{e,t} - \gamma_e\big(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\big) = 0 \ (t < T), \quad \pi_{e,T} \in \partial b_e$	(24)
(f) SOC limits, dynamics:	(K11) and (2)	(25)
(g) Network, flows:	$-\bar{f}_\ell \le f_{\ell,t} \le \bar{f}_\ell \ \perp \ \kappa_{\ell,t} - \big(A^\top \lambda_t\big)_\ell$	(26)
(h) Network, angles:	$A\,\mathrm{diag}(y)\,\kappa_t = 0, \qquad \theta_{r,t} = 0, \qquad f_{\ell,t} = y_\ell\big(A^\top \theta_t\big)_\ell$	(27)
(i) Market clearing:	(D1) $\perp$ $\lambda_{n,t}$ free	(28)

Definition 3 (Competitive equilibrium)**.** $(q^*, d^*, f^*, \theta^*, p^*, g^*, s^*, \lambda^*)$ is a competitive equilibrium if q_i^* solves (16) for every i, d_m^* solves (17) for every m, (p_e^*, g_e^*, s_e^*) solves (18) for every e, (f^*, θ^*) solves (19), all at prices λ^*, and the nodal balance (D1) holds.

7 Equivalence and the equilibrium theorem

Theorem 4 (Equivalence of the KKT system and the MCP)**.** *Let Assumptions 2 and 3 hold. Then z satisfies the ISO KKT system* (K1a)–(K12) *together with primal feasibility* (D1)–(D13) *if and only if z solves the MCP of Definition 2.*

Proof. ($\Rightarrow$) Partition the KKT conditions by the variable each is stationary in. (K1a) with (K8)–(K9) is (20), by the box-complementarity equivalence (K2′)–(9) applied to q; (K1b) with (K10) is (21); (K2) with (K12) is (22); (K3) with (K12) is (23); (K4)–(K5) is (24); (K11) and (D9) give (25); (K1c) with (K7) is (26); (K1d) with $\xi_t = 0$, together with primal feasibility of (D2) and (D4), is (27); and (D1) is (28). Every KKT condition is used exactly once and none is left over, because the Lagrangian separates as in (15).

($\Leftarrow$) Each step above is an identity between the same expressions, so the argument reverses term by term: the union of the agents' first-order conditions and market clearing reconstructs (K1a)–(K12) with the same multipliers. □

The content of Theorem 4 is that the ISO's KKT system is not merely *consistent* with the participants' conditions — it is *partitioned* by them. The blocks are disjoint and exhaustive, and the only condition belonging to no participant is market clearing, which is the ISO's own.

Theorem 5 (Equilibrium)**.** *Let Assumptions 1–4 hold and let* $(q^*, d^*, f^*, \theta^*, p^*, g^*, s^*)$ *solve* (5) *with multipliers* $\lambda^*, \ldots$ *Then it is a competitive equilibrium in the sense of Definition 3. Conversely, every competitive equilibrium yields an optimal solution of* (5)*.*

Proof. Under Assumption 1 each $-b_e$ is convex, so (5) minimises a convex objective; under Assumption 2 all constraints (D1)–(D13) are affine, so Abadie's constraint qualification holds automatically and KKT are necessary and sufficient for global optimality — for (5) and for each agent problem (16), (17), (18), (19), all of which are convex programs over polyhedra. Multipliers exist because the problem is feasible with finite optimal value. By Theorem 4, which uses Assumption 3, the ISO's optimality conditions are exactly the union of the agents' optimality conditions and market clearing, so sufficiency in each agent's problem gives global optimality of that agent's dispatch at λ^*, which is Definition 3; Assumption 4 guarantees that the problem each agent solves is the one the ISO solved on its behalf and that λ^* is its dual. The converse follows by reading Theorem 4 in the ($\Leftarrow$) direction and invoking sufficiency for (5). □

Corollary 6 (No profitable deviation for storage)**.** *At* λ^* *no storage owner can increase its surplus by deviating, even though it submits no in-horizon energy offer: its intertemporal opportunity cost is represented in the clearing by* π_e^* *through* (12)*, not by a price forecast.*

Corollary 7 (Individual rationality)**.** *The optimal value of* (18) *is non-negative:* $(p_e, g_e, s_{e,t}) = (0, 0, \gamma_e^t s_{e,0})$ *is feasible and yields* $b_e(\gamma_e^T s_{e,0} - \bar{s}_{e,T})$*, which is* 0 *when* $\bar{s}_{e,T} = \gamma_e^T s_{e,0}$*.*

Corollary 8 (Revenue adequacy)**.** *The merchandising surplus equals congestion rent and is non-negative:*

$$\mathrm{MS} := \Delta T \sum_{n,t} \lambda_{n,t} \Big[\textstyle\sum_{m,j} d_{m,j,t} + D_{n,t} - \sum_{i,k} q_{i,k,t} - \sum_e (g_{e,t} - p_{e,t}) \Big] = \Delta T \textstyle\sum_{\ell,t} \bar{f}_\ell \big(\bar{\mu}_{\ell,t} + \underline{\mu}_{\ell,t} \big) \ \geq 0. \tag{29}$$

Proof. By (D1) the bracket equals $-\sum_\ell A_{n\ell} f_{\ell,t}$, so $\mathrm{MS} = -\Delta T \sum_t f_t^\top A^\top \lambda_t = -\Delta T \sum_t f_t^\top (\kappa_t - \nu_t)$ by (K1c). Now $f_t^\top \kappa_t = \big(\mathrm{diag}(y) A^\top \theta_t\big)^\top \kappa_t = \theta_t^\top A \,\mathrm{diag}(y) \kappa_t = 0$ by (K1d), so $\mathrm{MS} = \Delta T \sum_t f_t^\top \nu_t = \Delta T \sum_{\ell,t} f_{\ell,t} (\bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t})$, and by (K7) $\bar{\mu}_{\ell,t} > 0$ only when $f_{\ell,t} = \bar{f}_\ell$ and $\underline{\mu}_{\ell,t} > 0$ only when $f_{\ell,t} = -\bar{f}_\ell$, giving (29). Note MS is exactly the optimal value of the network agent's problem (19). □

Remark 9 (What the equilibrium is an equilibrium in). Theorem 5 is an equilibrium in *revealed* parameters $(\alpha_e, \beta_e, \gamma_e, \epsilon_e, \zeta_e, \bar{s}_e, \underline{s}_e, w_e)$. It is an equilibrium in true values only if w_e equals the owner's true marginal value-to-go $\partial V_e / \partial s_{e,T}$. The design therefore relocates the exercise of market power from a price–quantity offer curve, which an ISO can screen against reference levels, to a declared value-to-go, which it generally cannot verify.

8 Hybrid participation: storage co-located with a variable resource

8.1 Two constructs that share a name

"Hybrid" denotes two arrangements with opposite formal properties, and most of the confusion in the design literature comes from not separating them.

1. **Co-optimised hybrid.** The ISO models the pair as a single resource: it holds the SOC variable, receives the end-of-horizon bid w_h, and enforces the shared interconnection limit itself. The owner submits parameters, not a schedule.
2. **Self-scheduled hybrid.** The ISO sees one meter and one net injection $z_{h,t}$; the internal split between plant output, charging and discharging is the owner's business. The owner submits a schedule, or a price–quantity curve on $z_{h,t}$.

Sections 8.2–8.6 treat the first: it preserves Assumptions 1–4 and every theorem of Section 7 goes through, but the storage's marginal valuation of energy changes character. Section 8.7 treats the second, which breaks Assumption 4 and, with it, the central claim of the SOC-based design.

8.2 The hybrid block

Let $\mathcal{H}$ be the set of hybrids, $h \in \mathcal{H}$ located at bus $n(h)$, consisting of a variable resource with availability $R_{h,t}$ and offer price c_h (usually 0) and a storage device with the storage parameters of Section 3, indexed by h in place of e. Write

$$z_{h,t} := r_{h,t} + g_{h,t} - p_{h,t} \tag{30}$$

for the net injection at the point of interconnection (POI). The objective (5) gains $\Delta T \sum_{h,t}\big[c_h r_{h,t} + \epsilon_h p_{h,t} + \zeta_h g_{h,t}\big] - \sum_h b_h(s_{h,T} - \bar{s}_{h,T})$, the nodal balance (D1) gains the term $-\sum_{h\in\mathcal{H}(n)} z_{h,t}$, constraints (D9)–(D13) are replicated with $e \to h$, and three new blocks appear:

$$r_{h,t} - R_{h,t} \le 0, \qquad -r_{h,t} \le 0, \qquad \Delta T\varphi_{h,t},\ \Delta T\underline{\varphi}_{h,t} \ge 0 \tag{D14}$$

$$z_{h,t} - \bar{P}_h \le 0, \qquad -z_{h,t} - \underline{P}_h \le 0, \qquad \Delta T\chi_{h,t},\ \Delta T\underline{\chi}_{h,t} \ge 0 \tag{D15}$$

$$p_{h,t} - r_{h,t} - \Theta_h \le 0, \qquad \Delta T\omega_{h,t} \ge 0 \tag{D16}$$

(D14) is availability with curtailment permitted; (D15) is the shared interconnection, in the injection and withdrawal directions; (D16) is the no-grid-charging rule imposed by several tax-credit regimes, written with a slack allowance $\Theta_h \ge 0$ so that its shadow value can be read off ($\Theta_h = 0$ is the strict rule, $\Theta_h = +\infty$ its absence). All three are affine, so Assumption 2 is untouched, and all three involve only variables of agent h, so Assumption 3 is untouched — provided the interconnection is not shared across owners (Section 8.8).

8.3 Stationarity of the hybrid block

Differentiating the augmented Lagrangian:

Plant output $r_{h,t}$.

$$\lambda_{n(h),t} = c_h + \varphi_{h,t} - \underline{\varphi}_{h,t} + \chi_{h,t} - \underline{\chi}_{h,t} - \omega_{h,t}. \tag{K1e}$$

Charging $p_{h,t}$.

$$\lambda_{n(h),t} + \epsilon_h + \omega_{h,t} - \chi_{h,t} + \underline{\chi}_{h,t} = \alpha_h\big(\pi_{h,t} - \bar{\nu}_{h,t}\big) + \underline{\rho}_{h,t} - \bar{\rho}_{h,t}. \tag{K2^h}$$

Discharging $g_{h,t}$.

$$\lambda_{n(h),t} - \chi_{h,t} + \underline{\chi}_{h,t} = \zeta_h + \beta_h\big(\pi_{h,t} + \underline{\nu}_{h,t}\big) - \underline{\sigma}_{h,t} + \bar{\sigma}_{h,t}. \tag{K3^h}$$

SOC. (D14)–(D16) contain no $s_{h,t}$, so (K4) and (K5) are reproduced *unchanged* with $e \to h$.

Define the **internal price** of the hybrid,

$$\boxed{\widetilde{\lambda}_{h,t} := \lambda_{n(h),t} - \chi_{h,t} + \underline{\chi}_{h,t}} \tag{31}$$

the price of a marginal MWh *behind* the point of interconnection. In terms of $\widetilde{\lambda}$, (K1e)–(K3^h) read

$$\text{plant:} \quad \widetilde{\lambda}_{h,t} = c_h + \varphi_{h,t} - \underline{\varphi}_{h,t} - \omega_{h,t}, \tag{32}$$

$$\text{charge:} \quad \widetilde{\lambda}_{h,t} + \epsilon_h + \omega_{h,t} = \alpha_h\big(\pi_{h,t} - \bar{\nu}_{h,t}\big) + \underline{\rho}_{h,t} - \bar{\rho}_{h,t}, \tag{33}$$

$$\text{discharge:} \quad \widetilde{\lambda}_{h,t} = \zeta_h + \beta_h\big(\pi_{h,t} + \underline{\nu}_{h,t}\big) - \underline{\sigma}_{h,t} + \bar{\sigma}_{h,t}. \tag{34}$$

Proposition 10 (Internal-price invariance)**.** *The hybrid storage block of the ISO KKT system is obtained from the standalone storage block* (K2)–(K5), (K11)–(K12) *by the single substitution* $\lambda_{n(e),t} \mapsto \widetilde{\lambda}_{h,t}$*, together with the addition of* $\omega_{h,t}$ *on the charging side. Consequently:*

1. *the Bellman recursion* (11) *and its solution* (12) *hold verbatim for* $\pi_{h,t}$*;*
2. *the reduced forms* (K2′)–(K3′) *hold with* $F^p_{h,t} = \widetilde{\lambda}_{h,t} + \epsilon_h + \omega_{h,t} - \alpha_h(\pi_{h,t} - \bar{\nu}_{h,t})$ *and* $F^g_{h,t} = \zeta_h + \beta_h(\pi_{h,t} + \underline{\nu}_{h,t}) - \widetilde{\lambda}_{h,t}$*;*
3. *simultaneous charging and discharging still never occurs at equilibrium;*
4. *the arbitrage condition* (13) *holds with* $\lambda \to \widetilde{\lambda}$ *and an extra* $\omega_{h,t}/\alpha_h$ *on the left-hand side, so that under* $\gamma_h = 1$ *and no binding SOC bound on* $(t, t']$*,*

$$\widetilde{\lambda}_{h,t'} = \frac{1}{\eta^h_{\mathrm{RT}}}\big(\widetilde{\lambda}_{h,t} + \epsilon_h + \omega_{h,t}\big) + \zeta_h. \tag{35}$$

Proof. (i) and (ii) are immediate from (33)–(34) and the fact that (K4)–(K5) are unchanged. (iii): suppose $p_{h,t}$ and $g_{h,t}$ are both strictly interior. Subtracting (34) from (33) eliminates $\widetilde{\lambda}_{h,t}$ and gives $\epsilon_h + \zeta_h + \omega_{h,t} + \alpha_h\bar{\nu}_{h,t} + \beta_h\underline{\nu}_{h,t} = (\alpha_h - \beta_h)\pi_{h,t}$. The left side is non-negative; the right side is non-positive whenever $\pi_{h,t} \geq 0$ and $\beta_h \geq \alpha_h$. Both vanish only in the knife-edge $\epsilon_h = \zeta_h = \omega_{h,t} = 0$, $\alpha_h = \beta_h$, in which case the two actions are payoff-identical. Note that (D16), far from creating simultaneity, makes it strictly harder by adding $\omega_{h,t}$ to the left side. (iv): eliminate π between (33) and (34) using (11) exactly as in (13). □

Proposition 10 is the reason hybridisation is far less destructive than it is often assumed to be: nothing in the intertemporal machinery of Section 5 depends on *which* price the storage faces, only on the fact that it faces one. What changes is the price itself, and that change is not innocuous.

8.4 Where the nodal price goes

Four regimes exhaust the possibilities, and only the first reproduces the standalone model.

Table 1: Regimes of the internal price. Entries assume $\underline{\chi}_{h,t} = 0$ (no binding withdrawal limit).

regime	binding	multipliers	internal price $\widetilde{\lambda}_{h,t}$
(i) POI slack	—	$\chi = \omega = 0$	$\lambda_{n(h),t}$ (standalone)
(ii) POI binding, plant at availability	(D15),(D14)	$\chi > 0, \varphi > 0$	$c_h + \varphi_{h,t}$
(iii) POI binding, plant curtailed	(D15)	$\chi > 0, \varphi = 0$	c_h
(iv) grid-charging rule binding	(D16)	$\omega > 0$	$c_h + \varphi_{h,t} - \omega_{h,t}$

Proposition 11 (Price decoupling under curtailment). *Suppose the hybrid curtails in interval t, i.e. $0 < r_{h,t} < R_{h,t}$, and that $\Theta_h = +\infty$ (no grid-charging rule) and $\underline{\chi}_{h,t} = 0$. Then $\varphi_{h,t} = \underline{\varphi}_{h,t} = 0$,* (K1e) *gives*

$$\chi_{h,t} = \lambda_{n(h),t} - c_h, \qquad \widetilde{\lambda}_{h,t} = c_h, \tag{36}$$

and (33) *reduces, for interior charging, to*

$$\boxed{\alpha_h\big(\pi_{h,t} - \bar{\nu}_{h,t}\big) = c_h + \epsilon_h} \tag{37}$$

in which the nodal price $\lambda_{n(h),t}$ does not appear.

This is the precise sense in which "λ disappears". It does not vanish from the market — loads still pay it and the thermal fleet is still dispatched against it — it vanishes from the *storage's* first-order condition, because the interconnection multiplier $\chi_{h,t}$ absorbs it exactly. The economics is transparent: the marginal MWh entering a curtailing hybrid's battery is not bought from the market at λ; it is diverted from spill, and the marginal social value of spilled energy is c_h, normally zero. A standalone storage at the same bus in the same interval must pay $\lambda_{n(h),t}$ for that MWh. Two identical batteries at one node therefore face charging costs that differ by the full nodal price, and this is not an arbitrage failure: they are consuming different goods.

Two consequences are worth isolating.

Corollary 12 (Degradation-threshold discharge). *Under the hypotheses of Proposition 11 with $c_h = 0$, $\gamma_h = 1$, and no SOC bound binding on $(t, t']$, the internal price at which the hybrid is willing to discharge is*

$$\widetilde{\lambda}_{h,t'} = \frac{\beta_h \epsilon_h}{\alpha_h} + \zeta_h, \tag{38}$$

a pure degradation threshold independent of the price at which the energy was acquired. For the parameters of Case C below ($\alpha = 0.9$, $\beta = 1$, $\epsilon = \zeta = 1$) this is \$2.11/MWh, against $(\lambda_{n(e),t} + 1)/0.9 + 1$ for a standalone.

Corollary 13 (Withheld discharge at a saturated POI). *In regime (ii) with $\chi_{h,t} > 0$,* (34) *shows the storage discharges only if $\lambda_{n(h),t} \geq \chi_{h,t} + \zeta_h + \beta_h(\pi_{h,t} + \underline{\nu}_{h,t})$. Since $\chi_{h,t}$ is largest exactly when the plant most wants the connection, a hybrid withholds discharge precisely in the intervals where λ is high, i.e. where the system values it most.*

Corollary 13 is the exact counterpart of the statement following Proposition 1. For a standalone unit the network's scarcity is *inside* $\lambda_{n(e),t}$, so no separate congestion term appears in the storage conditions. For a hybrid the interconnection's scarcity is *outside* $\lambda_{n(h),t}$ — there is no shift factor for a POI and it is not a row of $Cx \leq d$ — so a separate term $\chi_{h,t}$ does appear, and it works against the nodal signal rather than with it. Storage sited behind a saturated interconnection is, from the system's point of view, partly invisible.

8.5 What survives, and to whom the rents accrue

Theorem 14 (Equivalence and equilibrium with hybrids). *Let Assumptions 1–4 hold, with $\mathcal{H}$ a set of hybrids each wholly owned by one agent and each with its own interconnection. Then Theorems 4 and 5 hold with the MCP of Definition 2 augmented by*

$$\text{(j) Hybrid plant:} \qquad 0 \leq r_{h,t} \leq R_{h,t} \;\perp\; c_h + \chi_{h,t} - \underline{\chi}_{h,t} - \omega_{h,t} - \lambda_{n(h),t}, \tag{39}$$

$$\text{(k) Hybrid charge:} \qquad 0 \leq p_{h,t} \leq \bar{p}_h \;\perp\; \widetilde{\lambda}_{h,t} + \epsilon_h + \omega_{h,t} - \alpha_h(\pi_{h,t} - \bar{\nu}_{h,t}), \tag{40}$$

$$\text{(l) Hybrid discharge:} \qquad 0 \leq g_{h,t} \leq \bar{g}_h \;\perp\; \zeta_h + \beta_h(\pi_{h,t} + \underline{\nu}_{h,t}) - \widetilde{\lambda}_{h,t}, \tag{41}$$

$$\text{(m) POI and rule:} \qquad \text{(D15)} \;\perp\; \chi_{h,t},\ \underline{\chi}_{h,t} \geq 0, \qquad \text{(D16)} \;\perp\; \omega_{h,t} \geq 0, \tag{42}$$

together with (24)–(25) *for* h*, and with the hybrid's own problem*

$$\max_{r_h,p_h,g_h,s_h} \ \Delta T \sum_{t\in\mathcal{T}} \Big[\lambda_{n(h),t} z_{h,t} - c_h r_{h,t} - \epsilon_h p_{h,t} - \zeta_h g_{h,t}\Big] + b_h\big(s_{h,T} - \bar{s}_{h,T}\big) \tag{43}$$

subject to (2), (3), (D14)–(D16) *and the power bounds. Corollaries 6, 7 and 8 also hold unchanged.*

Proof. (D14)–(D16) are affine, so (5) remains a convex program over a polyhedron and Assumption 2 is preserved. They involve only variables indexed by h, so the decomposition (15) acquires one further additive block $\mathcal{L}_h$ and Assumption 3 is preserved; the block is, up to sign, (43). The partition argument of Theorem 4 then extends by three rows, (39)–(42), each used exactly once. For Corollary 7, $(r_h, p_h, g_h) = (0,0,0)$ with $s_{h,t} = \gamma_h^t s_{h,0}$ is feasible for (43) because (D14)–(D16) are all satisfied at zero. For Corollary 8, the proof uses only (D1), (K1c), (K1d) and (K7); hybrid injections enter the bracket of (29) exactly as storage injections do and cancel against (D1) in the same step. □

Remark 15 (Where the interconnection rent goes). Corollary 8 is unchanged, and that is itself the interesting fact. The POI is congested — $\chi_{h,t} > 0$ — yet the merchandising surplus contains no term for it. The rent

$$\mathrm{IR}_h := \Delta T \sum_{t\in\mathcal{T}} \chi_{h,t}\, \bar{P}_h \tag{44}$$

is collected by nobody in the settlement: it is an inframarginal rent internal to agent h, the difference between what the hybrid's output is worth at λ and what it is worth at $\widetilde{\lambda}$. It is the exact private analogue of congestion rent — same mathematics, different owner — and it is one reason hybridisation is privately attractive even where it is socially second best. It is also the quantity a shared-interconnection agreement must allocate (Section 8.8).

8.6 Social welfare

Let $W^\star$ denote the optimal value of the welfare problem (i.e. minus the optimal value of (5)). Because (D14)–(D16) enter linearly with the stated multipliers, the envelope theorem gives three exact sensitivities:

$$\frac{\partial W^\star}{\partial \bar{P}_h} = \Delta T \sum_{t\in\mathcal{T}} \chi_{h,t} \ \geq 0, \qquad \left.\frac{\partial W^\star}{\partial \Theta_h}\right|_{\Theta_h=0} = \Delta T \sum_{t\in\mathcal{T}} \omega_{h,t} \ \geq 0, \qquad \frac{\partial W^\star}{\partial \bar{s}_h} = \Delta T \sum_{t\in\mathcal{T}} \bar{\nu}_{h,t} \ \geq 0. \tag{45}$$

The first is the marginal social value of interconnection capacity, the second the marginal social cost of the no-grid-charging rule, the third the marginal social value of energy capacity. All three are verified numerically in Section 9.3.

The right counterfactual. It is tempting to conclude from (D14)–(D16) being a restriction of the feasible set that hybridisation destroys welfare. That comparison is against the wrong benchmark. Write

- W^{unc}: the plant and the storage each connected without limit;
- W^{hyb}: the pair sharing one interconnection of capacity $\bar{P}_h$;
- W^{sep}: the plant alone connected at $\bar{P}_h$, the storage separately and fully connected elsewhere.

Then $W^{\mathrm{unc}} \geq W^{\mathrm{hyb}}$ always, by feasible-set inclusion. But W^{hyb} versus W^{sep} is *ambiguous*, and it is the comparison that a queue-reform decision actually faces. Relative to W^{sep}, hybridisation

- **gains** the ability to convert energy that would otherwise be spilled — $\max\{0, R_{h,t} - \bar{P}_h\}$ MW of it — into stored energy at internal marginal cost $(c_h + \epsilon_h)/\alpha_h$ instead of $(\lambda_{n(h),t} + \epsilon_h)/\alpha_h$, by Proposition 11;
- **loses** the ability to discharge into the intervals the system values most, by Corollary 13.

Which effect dominates depends on the correlation between the plant's availability and the system's scarcity. For solar with an evening net-load peak the two are anti-correlated and the first effect tends to dominate; in Case C below $W^{\text{hyb}} = \$17{,}788.89$ against $W^{\text{sep}} = \$15{,}810.00$, a gain of \$1,978.89, while $W^{\text{unc}} = \$24{,}188.89$. The correct policy statement is therefore not that hybridisation is inefficient but that it **substitutes an operating restriction for an investment**: it is efficient exactly when

$$\Delta T \sum_{t \in \mathcal{T}} \chi_{h,t} \;\leq\; \text{annualised cost of one MW of interconnection,} \tag{46}$$

the left side being the first sensitivity in (45). In Case C this is \$160/MW over four hours.

Distributional and operational side effects. Three follow from Proposition 11 and are visible in Case C.

1. *The hybrid's SOC fills earlier.* With $\widetilde{\lambda}_{h,t} = c_h = 0$ the charging condition (37) is satisfied at $\pi_{h,t} - \bar{\nu}_{h,t} = \epsilon_h/\alpha_h$, a very low reservation value, so the hybrid charges whenever it is curtailing, regardless of the level of λ. The energy-capacity constraint (D10) therefore binds earlier than it would for a standalone: in Case C, $\bar{\nu}_{h,2} = \$37.89$/MWh.

2. *The costate can move against the price.* (37) pins $\pi_{h,t}$ from below by the internal cost of energy, not by λ. Case D shows the mirror image: imposing the grid-charging rule removes the outside option and raises the costate from \$45.56/MWh (set by $(\lambda + \epsilon)/\alpha$) to \$119/MWh (set by internal scarcity). A constraint intended to restrict the resource *raises* the shadow price of its stored energy.

3. *Consumers are unaffected at the margin, owners are not.* Because $\chi_{h,t}$ is a private multiplier, the settlement prices λ are formed by (K1a), (K1b) and Proposition 1 exactly as before. Load payments and generator revenues are unchanged in form; the entire effect of hybridisation on surplus is absorbed by the hybrid owner through IR_h and through the discharge withheld in Corollary 13. In Case C the whole \$4,800 welfare loss from a mis-forecast self-schedule is borne by the owner and none of it by consumers, because λ does not move; that is an artefact of the instance, not a theorem, but it is the typical direction.

8.7 The self-scheduled hybrid: what actually breaks

Now the second construct of Section 8.1. Suppose the ISO receives only $z_{h,t}$, either as a fixed schedule or as a price–quantity curve on net injection, and the internal split is decided by the owner. Then $s_{h,t}$, $\pi_{h,t}$, w_h and the whole of (D9)–(D16) disappear from the ISO's problem. Four things follow, in increasing order of severity.

(a) The costate is not computed. There is no SOC balance in the ISO's problem, hence no dual on it. The recursion (11) is not run, and w_h — the one scalar function the SOC-based design asks the participant for — has no terminal condition to supply. The design's central claim, Corollary 6, is not merely weakened; its hypothesis is absent.

(b) The price forecast returns. The owner must solve (43) at a forecast $\hat{\lambda}$. Let $V(z_h)$ be the optimal welfare of the rest of the system as a function of the hybrid's net injection profile, a concave piecewise-linear function under Assumptions 1–2, and let z_h^{ss} be the owner's best response to $\hat{\lambda}$. The welfare loss is

$$L_h := \max_z V(z) - V\big(z_h^{\text{ss}}\big) \;\geq 0, \qquad L_h = 0 \iff \hat{\lambda}_{n(h),\cdot} \in \partial V\big(z_h^{\text{ss}}\big). \tag{47}$$

Two readings. Under exact forecasts and price-taking behaviour, $L_h = 0$: self-scheduling is harmless in the rational-expectations benchmark, and it is worth saying so, because it explains why the

practice survives. But V is piecewise linear, so once the forecast error moves z_h^{ss} across a breakpoint of V the loss is *first order* in the error, not second order as differentiable intuition suggests. In Case C, an owner who under-forecasts the peak by 100 MW sees a flat price vector, holds its SOC one interval too long, and loses 27.0% of welfare.

(c) Price determinacy is destroyed. This is the least obvious consequence and the most damaging for the argument of Section 9.1. The observation following (9) — that a strictly interior resource pins λ at its bus — is what makes the dual of Case A unique and the published price vector infeasible. A self-scheduled hybrid contributes *no* complementarity condition: a fixed injection is a datum of the clearing problem, not a variable of it. The dual set at bus $n(h)$ is therefore enlarged, and the clearing range returns. Self-scheduling thus recreates precisely the indeterminacy that Section 9.1 identifies as the source of a 64% misstatement of storage surplus. The two failure modes are not independent: self-scheduling manufactures the conditions under which a non-dual price vector looks defensible.

(d) The revealed-parameter set widens. Remark 9 extends unfavourably. The parameters the ISO must take on trust grow from $(\alpha, \beta, \gamma, \epsilon, \zeta, \bar{s}, \underline{s}, w)$ to include $R_{h,t}$ and, in practice, $\bar{P}_h$. Of these, $R_{h,t}$ is a private meteorological forecast; understating it manufactures a synthetic $\varphi_{h,t}$ and, through (K1e), moves the hybrid's own nodal price in a direction a standalone storage cannot. Worse, w_h now conflates a value-to-go with an availability forecast, since the value of a stored MWh at T depends on what the co-located plant is expected to produce after T. An ISO can screen an offer curve against a reference level and can, with more difficulty, screen a value-to-go against historical price distributions; it cannot screen a joint declaration of value-to-go and private weather.

8.8 Shared interconnection under separate ownership

Suppose a set $\mathcal{S} \subseteq \mathcal{H}$ of resources under *different* ownership shares one interconnection, so that (D15) is replaced by

$$\sum_{h \in \mathcal{S}} z_{h,t} - \bar{P}_{\mathcal{S}} \leq 0, \qquad \Delta T \chi_{\mathcal{S},t} \geq 0. \tag{D15$'$}$$

The constraint is still affine, so Assumption 2 holds and the ISO problem still has a unique optimal value with well-defined duals. But it couples variables of distinct agents, so Assumption 3 fails and (15) no longer separates. Theorem 4 fails — not by non-convexity, and not by any defect the KKT conditions can detect, but by non-separability. This is a distinct failure mode from every entry in the source paper's taxonomy.

The structure is exactly that of Section 6.5. One may define a fictitious price-taking *interconnection agent* that buys the members' output at the internal price and sells it at the nodal price,

$$\max_{z} \ \Delta T \sum_{t \in \mathcal{T}} \big(\lambda_{n,t} - \widetilde{\lambda}_t\big) \sum_{h \in \mathcal{S}} z_{h,t} \quad \text{s.t.} \quad \sum_{h \in \mathcal{S}} z_{h,t} \leq \bar{P}_{\mathcal{S}}, \tag{48}$$

whose stationarity condition is $\lambda_{n,t} - \widetilde{\lambda}_t = \chi_{\mathcal{S},t}$, i.e. (31), and whose optimal value is $\mathrm{IR}_{\mathcal{S}}$ of (44). With a single owner this device is redundant: the owner *is* the agent, and the rent is internal. With several owners it is not redundant, and the model closes only if the members trade the shared capacity at $\chi_{\mathcal{S},t}$ — that is, only if there is an internal market at the POI. In its absence the capacity is allocated by an administrative rule (pro rata, priority order, first-scheduled), the members' effective internal prices differ from $\chi_{\mathcal{S},t}$, and the dispatch that decentralises is not the ISO's. The parallel with loop flow in Section 6.5 is exact: there, too, a multiplier exists that no nodal price supports, and there, too, the fix is to hand the constrained object to a single decision maker.

8.9 Non-convex and horizon-coupling hybrid features

Three further features deserve classification.

- *Shared inverter.* A single AC-coupled inverter serving both the plant and the discharge path gives $r_{h,t}+g_{h,t} \le \bar{P}^{\text{inv}}_h$, which is affine and therefore harmless; it merely adds a second multiplier of the same type as $\chi_{h,t}$. Minimum inverter loading, or a rule that the plant and the battery must use the connection in disjoint windows, introduces binaries and breaks Assumption 2 in exactly the way Appendices A.2–A.3 of [1] do.
- *DC coupling.* A DC-coupled hybrid in which charging can only occur from the plant's DC bus is (D16) with $\Theta_h = 0$ imposed by physics rather than by regulation. It is affine, hence A2-safe; its cost is $\Delta T \sum_t \omega_{h,t}$ by (45), quantified in Case D as \$403.94 on a \$11,944.44 welfare base.
- *Annual tax-credit charging limits.* A rule of the form $\sum_{t\in\text{year}} \max\{0, p_{h,t} - r_{h,t}\} \le \varsigma \sum_{t\in\text{year}} p_{h,t}$ is convex for $\varsigma \in [0,1]$ and therefore preserves Assumptions 1–2. Its difficulty is elsewhere: it couples intervals *beyond* the clearing horizon T. It therefore needs its own costate and its own terminal condition — a second value-to-go bid, declared on cumulative grid-charged energy rather than on SOC. Without one, the day-ahead problem prices the SOC scarcity and leaves the annual scarcity unpriced, which is the same pathology the last bullet of Section 10 describes for a rolling horizon, applied to a different state variable.

9 Numerical verification

9.1 Case A: the single-node base case of [1]

Section 4.7 of [1] is the specialisation $|\mathcal{N}| = 1$, $\mathcal{L} = \emptyset$ (hence $\Psi = 0$ and all $\lambda_{n,t} = \lambda_t$), $|\mathcal{E}| = 1$, no ramp limits, inelastic demand. Data: $T = 6$, $\Delta T = 4$ h, $\alpha = 0.9$, $\beta = \gamma = 1$, $\epsilon = \zeta = 1$ \$/MWh, $\bar{s} = 800$ MWh, $\underline{s} = 0$, $s_0 = 0$, $\bar{s}_T = 0$, $w = 40$ \$/MWh; demand $(100, 200, 300, 300, 200, 100)$ MW; renewable availability $(200, 100, 500, 500, 25, 0)$ MW at \$0; thermal capacity 100 MW at \$50/MWh. Solving (5) as an LP reproduces the published dispatch, with the charging split between intervals 3 and 4 degenerate (both splits give objective \$19,448.89).

Table 2: Case A: optimal duals (\$/MWh) and dispatch.

t	1	2	3	4	5	6
π_t	49.000	1.111	1.111	1.111	49.000	40.000
$\bar{\nu}_t$	0	0	0	0	47.889	0
$\underline{\nu}_t$	0	47.889	0	0	0	9.000
λ_t	**43.100**	50.000	0	0	50.000	**50.000**
p_t (MW)	100	0	22.2	200	0	0
g_t (MW)	0	90	0	0	100	100
s_t (MWh)	360	0	80	800	400	0

Every condition is satisfied: (K5) $\pi_6 = w = 40$; (K4) $\pi_5 = 40 - 0 + 9 = 49$ and $\pi_4 = 49 - 47.889 = 1.111$; (K2′) at $t = 1$ (interior charging) $\lambda_1 + \epsilon = 44.1 = 0.9 \times 49$, at $t = 3, 4$ $\lambda_t + \epsilon = 1 = 0.9 \times 1.111$; (K3′) at $t = 2$ (interior discharging) $\lambda_2 = 1 + 1.111 + 47.889 = 50$, at $t = 5$ $1 + 49 + 0 = 50$, at $t = 6$ $1 + 40 + 9 = 50$; the idle-direction inequalities hold strictly ($F^p_t = 50, 50, 15 > 0$ at $t = 2, 5, 6$ and $F^g_t = 6.9, 2.111, 2.111 > 0$ at $t = 1, 3, 4$); and (13) for $t = 4 \to t' = 5$ gives $1.111 = 49 - 47.889$, the simplified form (14) not applying because $\bar{\nu}_5 > 0$.

The published price vector is not dual feasible. Reference [1] reports $\lambda = (0, 50, 0, 0, 50, 41)$, arguing that any price in $[0, 43.1]$ clears interval 1 and any price in $[41, 50]$ clears interval 6. That argument tests clearing only against the *offering* resources: renewable at its cap accepts any $\lambda \ge 0$,

thermal at zero output accepts any $\lambda \le 50$. Because the storage submits no in-horizon bid it drops out of that test — but (K2′) applies to it all the same, and $p_1 = 100$ MW is strictly interior, so $\lambda_1 = \alpha\pi_1 - \epsilon = 0.9 \times 49 - 1 = 43.1$ exactly; similarly (K3′)–(K4) force $\lambda_5 = \lambda_6 = 50$. There is no clearing range: perturbing each interval's demand by $\pm 10^{-3}$ MW gives identical left and right derivatives of the optimal value in all six intervals $(43.100, 50.000, 0, 0, 50.000, 50.000)$, so the dual is unique. Note that this argument depends on the storage being a *variable* of the clearing problem; Section 8.7(c) shows that it collapses the moment the resource self-schedules.

Table 3: Case A: storage surplus, best response versus ISO dispatch.

price vector λ	best-response surplus	surplus at ISO dispatch
$(0, 50, 0, 0, 50, 41)$ (as published)	\$76,622	\$51,951
$(43.1, 50, 0, 0, 50, 50)$ (dual feasible)	\$38,311	\$38,311

At the published prices the storage strictly prefers to deviate: it charges 222 MW in interval 1 rather than 100 — energy is free at $\lambda_1 = 0$ — and again in interval 3, discharging 200 MW in intervals 2 and 5. The gap of \$24,671 is a genuine equilibrium failure. At the dual-feasible prices the ISO dispatch attains the maximum exactly, as Theorem 5 requires.

9.2 Case B: three buses, four intervals, with congestion and ramping

To exercise (K1a)–(K1d) and Proposition 1 we solve a networked case. Buses $\{1, 2, 3\}$, lines $\{(1,2), (2,3), (1,3)\}$ with $y_\ell = 1$ and $\bar{f}_\ell = 100$ MW, reference $r = 1$; $T = 4$, $\Delta T = 1$ h. Unit 1 at bus 1 offers two blocks, 100 MW at \$20 and 100 MW at \$40, ramp ± 40 MW; unit 2 at bus 3 offers 300 MW at \$90 with ramp ± 50 MW. Fixed demand $(20, 25, 30, 20)$ MW at bus 2 and $(80, 120, 215, 90)$ MW at bus 3; one elastic block of 40 MW at \$100 at bus 3. One storage unit at bus 3: $\alpha = 0.9$, $\beta = \gamma = 1$, $\epsilon = \zeta = 1$, $\bar{s} = 20$ MWh, $\underline{s} = 0$, $s_0 = \bar{s}_T = 0$, $\bar{p} = \bar{g} = 60$ MW, $w = 30$ \$/MWh. Optimal value \$13,667.22.

Table 4: Case B: primal solution and multipliers.

t	1	2	3	4
$\lambda_{1,t}$ (\$/MWh)	40	40	40	40
$\lambda_{2,t}$	65	65	90	40
$\lambda_{3,t}$	90	90	**140**	40
$\bar{\mu}_{(1,3),t}$	75	75	150	0
$\kappa_{(1,2),t} = \kappa_{(2,3),t}$	25	25	50	0
$\kappa_{(1,3),t}$	-25	-25	-50	0
$q_{1,1,t}$ / $q_{1,2,t}$ (MW)	100 / 60	100 / 62.5	100 / 65	100 / 40
$q_{2,1,t}$ (MW)	0	24.722	60	10
$d_{1,1,t}$ (MW)	40	40	0	40
$f_{(1,3),t}$ (MW)	100	100	100	86.667
p_t / g_t (MW)	20 / 0	2.222 / 0	0 / 20	0 / 0
s_t (MWh)	18	20	0	0
π_t (\$/MWh)	101.111	101.111	139.000	30.000
$\bar{\nu}_t$ / $\underline{\nu}_t$	0 / 0	0 / 0	37.889 / 0	0 / 109

Checks, interval by interval:

- **(K1a), unit 1.** Block 1 is at its cap in every interval, $\bar{\delta}_{1,1,t} = 20$, and block 2 is interior, so $\lambda_{1,t} = c_{1,2} = \$40$ for all t: the marginal block prices the reference bus and no ramp binds ($\varrho_{1,t} = 0$).
- **(K1a), unit 2, and the ramp wedge.** At $t = 3$ unit 2 is interior ($q = 60$, cap 300, ramp slack), so $\lambda_{3,3} = c_{2,1} + \varrho_{2,3}$. The down-ramp at $t = 4$ binds, $\tau_{2,4} = 50$, giving $\varrho_{2,3} =$

$-(\bar{\tau}_{2,4} - \underline{\tau}_{2,4}) = +50$ and $\lambda_{3,3} = 90 + 50 = \$140$. A binding *future* down-ramp raises the *present* price by exactly the multiplier. At $t = 4$, $q = 10$ with $\varrho_{2,4} = -50$ and $\underline{\delta} = 0$: $\lambda_{3,4} = 90 - 50 = \$40$.

- **(K1b).** The elastic block is served in full at $t = 1, 2, 4$ with $\bar{\psi} = 10, 10, 60$, giving $\lambda_{3,t} = 100 - \bar{\psi}$; at $t = 3$ it is curtailed to zero with $\underline{\psi} = 40$, giving $\lambda_{3,3} = 100 + 40 = \$140$.
- **(K1c)–(K1d).** $A\,\mathrm{diag}(y)\kappa_t = 0$ holds to machine precision in all four intervals, and $\xi_t = 0$ as predicted.
- **Proposition 1.** With $\Psi_{(1,3),2} = -1/3$ and $\Psi_{(1,3),3} = -2/3$, (10) gives at $t = 3$ $\lambda_2 = 40 + \frac{1}{3}(150) = \$90$ and $\lambda_3 = 40 + \frac{2}{3}(150) = \$140$; likewise $(65, 90)$ at $t = 1, 2$. All six congested-interval prices reproduce exactly.
- **(K2)–(K5).** $\pi_4 = w = 30$; $\pi_3 = 30 - 0 + 109 = 139$; $\pi_2 = 139 - 37.889 + 0 = 101.111 = \pi_1$. Charging is interior at $t = 1, 2$: $\lambda_{3,t} + \epsilon = 91 = 0.9 \times 101.111$. Discharging is interior at $t = 3$: $\lambda_{3,3} = 140 = 1 + 139 + 0$. Idle directions hold strictly.
- **Corollary 8.** MS = \$30,000, and $\sum_{\ell,t} \bar{f}_\ell(\bar{\mu} + \underline{\mu}) = 100 \times (75 + 75 + 150) = \$30{,}000$: identical.

Finally, the equilibrium property itself. At $\lambda_{3,\cdot} = (90, 90, 140, 40)$ the storage's best response to (18) yields surplus \$757.78, exactly its surplus at the ISO dispatch; unit 2's best response to (16) yields \$2,500, again exactly its profit at the ISO dispatch. Both agents have multiple optima — the ISO dispatch is one of them — which is precisely what Theorem 5 asserts and is the generic situation when a resource is marginal.

9.3 Case C: a co-optimised hybrid at a saturated interconnection

To exercise (K1e)–(K3^h) and Propositions 10 and 11 we deliberately remove the network, so that every price difference in the case is attributable to the interconnection and not to congestion. One bus, $\mathcal{L} = \emptyset$; $T = 4$, $\Delta T = 1$ h. One thermal unit with blocks 200 MW at \$40 and 100 MW at \$120, no ramp limits. Fixed demand $(100, 100, 280, 150)$ MW and one elastic block of 50 MW at \$200. One hybrid at the bus: plant availability $R_h = (250, 250, 40, 0)$ MW at $c_h = 0$; $\bar{P}_h = 100$ MW, $\underline{P}_h = 100$ MW; $\alpha = 0.9$, $\beta = \gamma = 1$, $\epsilon = \zeta = 1$ \$/MWh, $\bar{s} = 100$ MWh, $\underline{s} = 0$, $s_0 = \bar{s}_T = 0$, $\bar{p} = \bar{g} = 100$ MW, $w = 30$ \$/MWh; no grid-charging rule ($\Theta_h = +\infty$). Optimal welfare \$17,788.89.

Table 5: Case C: primal solution, multipliers and the internal price.

t	1	2	3	4
$q_{1,1,t}$ / $q_{1,2,t}$ (MW)	50 / 0	50 / 0	200 / 30	160 / 0
$d_{1,1,t}$ (MW)	50	50	50	50
$r_{h,t}$ (MW)	200	111.111	40	0
$p_{h,t}$ / $g_{h,t}$ (MW)	100 / 0	11.111 / 0	0 / 60	0 / 40
$s_{h,t}$ (MWh)	90	100	40	0
$z_{h,t}$ (MW)	100	100	100	40
λ_t (\$/MWh)	40	40	**120**	40
$\chi_{h,t}$	40	40	80	0
$\varphi_{h,t}$	0	0	40	40
$\widetilde{\lambda}_{h,t} = \lambda_t - \chi_{h,t}$	**0**	**0**	40	40
$\pi_{h,t}$	1.111	39.000	39.000	30.000
$\bar{\nu}_{h,t}$ / $\underline{\nu}_{h,t}$	0 / 0	37.889 / 0	0 / 0	0 / 9

Checks:

- (K1a), (K1b). Block 1 is interior at $t = 1, 2, 4$, so $\lambda_t = \$40$; at $t = 3$ block 2 is interior and $\lambda_3 = \$120$. The elastic block is served in full throughout, with $\bar{\psi} = (160, 160, 80, 160)$.

- (K1e) **and Proposition 11.** At $t = 1, 2$ the plant is curtailed, $r_{h,t} < R_{h,t}$, so $\varphi_{h,t} = 0$ and $\lambda_t = \chi_{h,t} = \$40$: the interconnection multiplier absorbs the nodal price exactly, and $\widetilde{\lambda}_{h,1} = \widetilde{\lambda}_{h,2} = 0$. At $t = 3$ the plant is at availability, $\varphi_{h,3} = \$40$, and $\lambda_3 = \varphi_{h,3} + \chi_{h,3} = 40 + 80 = \$120$. (At $t = 4$, $R_{h,4} = 0$ places $r_{h,4}$ at both of its bounds, so the split of the multiplier between φ and $\underline{\varphi}$ is arbitrary; the identity that is pinned is $\varphi_{h,4} - \underline{\varphi}_{h,4} = \lambda_4 = \$40$.)
- (K4)–(K5). $\pi_{h,4} = w = 30$; $\pi_{h,3} = 30 - 0 + 9 = 39$; $\pi_{h,2} = 39 - 0 + 0 = 39$; $\pi_{h,1} = 39 - 37.889 = 1.111$. The recursion is identical in form to Case A, as Proposition 10(i) requires.
- (33) **and** (37). Charging is strictly interior at $t = 2$: $\widetilde{\lambda}_{h,2} + \epsilon = 0 + 1 = 1 = \alpha(\pi_{h,2} - \bar{\nu}_{h,2}) = 0.9 \times 1.111$. The reservation value $\pi_{h,2} - \bar{\nu}_{h,2} = 1/0.9 = \$1.111$/MWh is set entirely by degradation and efficiency, with no trace of $\lambda_2 = \$40$. At $t = 1$ charging is at its power bound and $F^p_{h,1} = 1 - 0.9 \times 1.111 = 0 \leq 0$; the split of charging between $t = 1$ and $t = 2$ is degenerate, exactly as in Case A.
- (34) **and Corollary 13.** Discharging is interior at $t = 3$: $\widetilde{\lambda}_{h,3} = 40 = \zeta + \beta(\pi_{h,3} + \underline{\nu}_{h,3}) = 1 + 39$. The nodal price is \$120 but the storage is remunerated internally at \$40, and it discharges only 60 MW of the 100 MW it could: the POI, not the market, sets the quantity. At $t = 4$, $\widetilde{\lambda}_{h,4} = 40 = 1 + 30 + 9$.
- **Envelope identities** (45). $\sum_t \chi_{h,t} = \$160$/MW and re-solving with $\bar{P}_h = 101$ MW gives welfare \$17,948.89, a difference of exactly \$160. $\sum_t \bar{\nu}_{h,t} = \$37.889$/MWh and re-solving with $\bar{s}_h = 101$ MWh gives \$17,826.78, a difference of exactly \$37.889.
- **Corollary 8.** With $\mathcal{L} = \emptyset$, MS $= 0$ in every interval, as (29) requires — while the interconnection rent $\mathrm{IR}_h = 100 \times (40 + 40 + 80) = \$16{,}000$ accrues entirely to the owner and appears nowhere in the settlement (Remark 15).

Counterfactuals. Re-solving the same data under the three arrangements of Section 8.6:

Table 6: Case C: welfare under the three interconnection arrangements.

arrangement		welfare	charging price seen by the storage
W^{unc}	no interconnection limit	\$24,188.89	$\lambda = (0, 0, 40, 40)$
W^{hyb}	shared POI, $\bar{P}_h = 100$ MW	\$17,788.89	$\widetilde{\lambda} = (0, 0, 40, 40)$
W^{sep}	plant capped at 100 MW, storage elsewhere	\$15,810.00	$\lambda = (40, 40, 46.56, 40)$

The ordering $W^{\mathrm{unc}} > W^{\mathrm{hyb}} > W^{\mathrm{sep}}$ is the point of the case. Hybridisation costs \$6,400.00 against an unconstrained build-out — the value of the interconnection the hybrid did not build — but gains \$1,978.89 against the arrangement in which the same plant is capped and the storage sits elsewhere, because the hybrid converts curtailed energy at an internal price of zero where the standalone must buy at \$40. In W^{sep} the standalone's costate is $\pi_1 = \pi_2 = 45.556 = (40 + 1)/0.9$, the textbook grid-charging value; in W^{hyb} it is $1.111 = 1/0.9$. The two numbers differ by exactly λ/α, which is (36).

Self-scheduling with a mis-forecast peak. Suppose the hybrid self-schedules (Section 8.7) against prices forecast from a market model in which the $t = 3$ demand is taken to be 180 MW rather than 280 MW. That model returns $\hat{\lambda} = (40, 40, 40, 40)$: with a flat price the owner sees no reason to release energy at $t = 3$ and holds it for $t = 4$. Its self-schedule is $z^{\mathrm{ss}}_h = (100, 100, 40, 100)$ MW against the co-optimised $z^{\star}_h = (100, 100, 100, 40)$ MW. Imposing z^{ss}_h on the clearing gives welfare \$12,988.89, a loss of \$4,800.00, or **27.0%**. The arithmetic is exact: 60 MWh are moved from an interval priced at \$120 to one priced at \$40, and $60 \times 80 = \$4{,}800$; the ISO must run the \$120 block for 60 MWh more at $t = 3$ and the \$40 block for 60 MWh less at $t = 4$. Because λ is unchanged, the whole loss falls on the owner, whose surplus drops from \$21,388.89 to

\$16,588.89 — the numerical illustration of (47) and of the third bullet of Section 8.6. Under the co-optimised design the same owner submits only $w_h = 30$ and never forms a forecast at all.

9.4 Case D: the cost of a no-grid-charging rule

Same bus, thermal offer, demand and elastic block as Case C, but with plant availability $R_h = (250, 5, 0, 0)$ MW and $\bar{s}_h = 300$ MWh, so that the storage has room for more energy than the plant can supply and grid charging becomes attractive. Two runs: without (D16) ($\Theta_h = +\infty$) and with the strict rule ($\Theta_h = 0$).

Table 7: Case D: with and without the no-grid-charging rule.

	$\Theta_h = +\infty$ (no rule)				$\Theta_h = 0$ (strict rule)			
t	1	2	3	4	1	2	3	4
$r_{h,t}$ (MW)	200	5	0	0	200	5	0	0
$p_{h,t}$ (MW)	100	11.111	0	0	100	5	0	0
$g_{h,t}$ (MW)	0	0	100	0	0	0	94.5	0
$z_{h,t}$ (MW)	100	−6.111	100	0	100	0	94.5	0
$s_{h,t}$ (MWh)	90	100	0	0	90	94.5	0	0
λ_t (\$/MWh)	40	40	120	40	40	40	120	40
$\chi_{h,t}$	40	0	0	0	40	0	0	0
$\varphi_{h,t}$	0	40	—	—	0	106.100	—	—
$\omega_{h,t}$	0	0	0	0	0	**66.100**	0	0
$\pi_{h,t}$	45.556	45.556	45.556	30	119	119	119	30
$\underline{\nu}_{h,t}$	0	0	0	15.556	0	0	0	89
welfare	\$11,944.44				\$11,540.50			

Checks and reading:

- Without the rule the hybrid *withdraws* at $t = 2$ ($z_{h,2} = -6.111$ MW) and its costate is $\pi_{h,t} = 45.556 = (\lambda_t + \epsilon)/\alpha$ throughout $t = 1, 2, 3$: the outside option of buying from the grid pins the value of stored energy at its acquisition cost.
- With the rule, $p_{h,2} = r_{h,2} = 5$ MW binds (D16) with $\omega_{h,2} = \$66.10$/MWh. Check (K1e): $\lambda_2 = \varphi_{h,2} + \chi_{h,2} - \omega_{h,2} = 106.1 + 0 - 66.1 = \$40$. Check (33) for interior charging: $\widetilde{\lambda}_{h,2} + \epsilon + \omega_{h,2} = 40 + 1 + 66.1 = 107.1 = \alpha\pi_{h,2} = 0.9 \times 119$. Check (34) for interior discharging at $t = 3$: $\widetilde{\lambda}_{h,3} = 120 = \zeta + \beta(\pi_{h,3} + \underline{\nu}_{h,3}) = 1 + 119$. Check (K4): $\pi_{h,3} = 30 - 0 + 89 = 119$.
- **The rule raises the costate.** Removing the outside option lifts $\pi_{h,t}$ from \$45.56 to \$119/MWh: with no way to replenish, the stored MWh is valued at what it can earn rather than at what it cost. This is the mirror image of Proposition 11, where curtailment pushes the costate *down* to ϵ/α. Both are instances of the same mechanism — π tracks the binding internal margin, whatever that margin happens to be — and in neither is the nodal price the relevant quantity.
- **Welfare.** The rule costs \$403.94, or 3.4% of welfare, on this instance, and $\partial W^\star/\partial\Theta_h = \Delta T \sum_t \omega_{h,t} = \$66.10$ per MW of allowance at the margin, confirming the second identity of (45).

Note that in Case C the same rule would have been slack: with $\bar{P}_h$ binding in every interval in which the plant produces, the hybrid never wants grid energy. The no-grid-charging rule and a saturated interconnection are substitutes, and the cost of the former is concentrated in exactly the hours the latter does not cover.

10 Where the equilibrium breaks

Table 8: Failure modes by assumption. A1: curvature of b_e. A2: convexity of the feasible set. A3: separability of non-clearing constraints by ownership. A4: congruence of the ISO's model and prices with the agent's.

feature	A1	A2	A3	A4
w_e increasing (invalid bid)	**fails**	—	—	—
mutual exclusivity via binaries (A.2 of [1])	—	**fails**	—	—
minimum charge/discharge levels (A.3 of [1])	—	**fails**	—	—
unit commitment, minimum stable levels	—	**fails**	—	—
AC network / non-convex losses	—	**fails**	—	—
power foldback (A.1.1 of [1])	—	—	—	—
SOC-dependent degradation epigraph (A.1.2 of [1])	—	—	—	—
co-optimised hybrid: POI, curtailment, DC coupling	—	—	—	—
annual tax-credit charging limit (within horizon)	—	—	—	—
shared inverter, minimum loading or disjoint windows	—	**fails**	—	—
shared POI, several owners (D15′)	—	—	**fails**	—
self-scheduled hybrid (no SOC visibility)	—	—	—	**fails**
ISO omits the POI the owner faces	—	—	—	**fails**
prices not duals of (5) (Case A)	—	—	—	**fails**

- **Non-concave bid.** If w_e increases anywhere, $-b_e$ is non-convex, (K5) is no longer sufficient, and no supporting price need exist. Monotonicity of w_e should be enforced as a bid-validity rule, exactly as monotone offer stacks are for thermal units.
- **Binaries.** With A.2/A.3, or with unit commitment on the thermal side, the feasible set is non-convex, KKT are only locally necessary, a duality gap opens and uplift is required. This is the same pathology as three-part offers, not a storage-specific one.
- **Network non-convexity.** (D2) is the DC linearisation. With a full AC model or non-convex loss representation, Assumption 2 fails, Proposition 1 loses its exactness and the network block of the MCP no longer matches the ISO's.
- **Convexity-preserving extensions.** Foldback is linear; the degradation epigraph $D_t \geq \max\{\cdot, \cdot\}$ is convex; piecewise-linear convex losses are fine; and, by Theorem 14, so is the whole of the co-optimised hybrid block (D14)–(D16). All preserve Theorems 4–5 unchanged.
- **Non-separability.** A shared interconnection under separate ownership, (D15′), is the first feature in this taxonomy that is convex and yet breaks the equivalence, because it couples the blocks of (15). No amount of convexification repairs it; only an assignment of the shared asset to a single decision maker, or an explicit internal market at the POI, does (Section 8.8).
- **Incongruence.** Three distinct constructs share this column. If the ISO clears a hybrid as two independent resources at one bus while the owner is bound by $\bar{P}_h$, the ISO's dispatch can be *infeasible* for the owner. If the resource self-schedules, the ISO's problem no longer contains the SOC state, the costate (11) is not computed, and the price determinacy of Section 4.3 is lost. And if the prices used in settlement are not duals of (5), as in Case A, the model is convex, an equilibrium exists, and it is simply not the one being settled. The three differ in how hard they are to repair: the first is a modelling omission, the second a market-design choice, the third a computational slip and the easiest of all to fix.
- **Real time.** Theorem 5 is stated for a fully financially binding horizon. If only $t = 1$ binds and the horizon rolls, the settlement participants face is not the one the KKT system prices, and the multi-interval pricing problem reappears; the day-ahead position then enters the real-time problem as the reference $\bar{s}_{e,T}$. The same remark applies, with a different state variable, to the annual charging limits of Section 8.9.

11 Conclusion

Written out in full, the ISO problem prices five distinct scarcities: fuel and opportunity cost through (K1a), willingness to pay through (K1b), transmission through (K1c)–(K1d) and Proposition 1, ramping through the wedge $\varrho_{i,t}$, and stored energy through the costate recursion (11). Its KKT system decomposes exactly into the complementarity conditions of the individual agents plus market clearing (Theorem 4); under a concave end-of-horizon bid, a convex feasible set, ownership separability and model congruence those conditions are necessary and sufficient, so the ISO dispatch with its duals is a competitive equilibrium supporting no profitable deviation (Theorem 5), storage surplus is non-negative, and merchandising surplus equals congestion rent (Corollary 8). The storage owner never needs a price forecast — but the ISO must use the duals of its own problem as prices, and Case A shows that this is not automatic: on the published base case the reported prices are not dual feasible, the dual is unique, and the resulting misstatement of storage surplus is 64%.

Hybridisation adds a sixth scarcity, the shared interconnection, and it is the only one of the six that is priced *outside* λ. The whole storage block survives the change under a single substitution, $\lambda_{n(h),t} \mapsto \widetilde{\lambda}_{h,t} = \lambda_{n(h),t} - \chi_{h,t}$ (Proposition 10): the Bellman recursion, the impossibility of simultaneous charge and discharge, individual rationality and revenue adequacy all carry over verbatim, and the co-optimised hybrid remains a competitive equilibrium (Theorem 14). What does not carry over is the connection between the storage's valuation of energy and the market's. When the hybrid curtails, $\chi_{h,t}$ absorbs $\lambda_{n(h),t}$ exactly and the nodal price leaves the storage's first-order condition altogether (Proposition 11), leaving a pure degradation threshold; when the interconnection saturates in a high-price interval, the same multiplier withholds discharge precisely when the system values it most (Corollary 13). Neither effect is a market failure — both are the correct shadow prices of a real constraint — but both mean that a MW of hybrid storage and a MW of standalone storage at the same node are not the same product, and the settlement cannot tell them apart, because IR_h never enters it (Remark 15).

The two genuinely damaging arrangements are elsewhere, and neither is detectable by convexity arguments. A shared interconnection under separate ownership breaks the separability that Theorem 4 rests on; the fix is institutional, not mathematical. And a self-scheduled hybrid removes the SOC state from the ISO's problem altogether, which discards the costate recursion, hands the price forecast back to the owner — 27.0% of welfare in Case C when the forecast is wrong — and, less obviously, restores the price indeterminacy that Case A shows to be worth 64% of storage surplus. The SOC-based design's central promise is that the ISO computes the intertemporal opportunity cost so the participant does not have to. Hybridisation is compatible with that promise; hybridisation behind a single meter is not.

References

[1] R. Baldick. State-of-charge based participation model for energy storage in ISO markets. Source manuscript; equation numbers (1)–(9), Tables 2–4, Sections 4.5–4.7 and Appendices A.1–A.4 cited in this note refer to it.